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FINITE SYMPLECTIC AUTOMORPHISMS OF HYPERKÄHLER MANIFOLDS OF $K3^{[n]}$-TYPE ACT TRIVIALLY ON DISCRIMINANT GROUPS

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Abstract. We give an alternative proof of Mongardi's result that every finite group of symplectic automorphisms of a hyperkähler manifold of $K3^{[n]}$-type acts trivially on the discriminant groups of its second cohomology lattice and its coinvariant lattice. The proof combines sphere-packing bounds, Vinberg's reflection chambers and structure results on even 2-elementary lattices.

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1. Introduction

The study of symplectic automorphisms of hyperkähler manifolds has been an active area of research in recent years. Let X be a hyperkähler manifold, and let $G \subset \mathrm{Aut}^s(X)$ be a finite group of symplectic automorphisms. Let

$$S = S_G(H^2(X,\mathbb{Z})) = (H^2(X,\mathbb{Z})^G)^\perp$$

be the coinvariant lattice for the induced action of G on $H^2(X,\mathbb{Z})$. The lattice S is always negative definite. If, moreover, the induced action of G on the discriminant group $A_{H^2(X,\mathbb{Z})}$ is trivial, then its action on A_S is also trivial. For X of $K3^{[n]}$-type, Mongardi [Mon16, Theorem 25] shows that $S(-1)$, obtained by reversing the sign of the pairing on S, embeds primitively into the positive-definite Leech lattice $\mathbb{L}$ as a coinvariant sublattice, so it is one of the 290 lattices classified by Höhn–Mason [HM16]. He also conjectures that $\mathrm{rk}(S)+\ell(A_S) < 24$ [Mon16, Conjecture 32], where $\ell(A_S)$ is the minimal number of generators of A_S. We prove this conjecture in a another paper [FWZ26].

For a hyperkähler manifold X of $K3^{[n]}$-type with $n \geq 2$, write $L_n = H^2(X, \mathbb{Z})$. Then $A_{L_n} \cong \mathbb{Z}/(2n-2)\mathbb{Z}$. Markman [Mar11, Lemma 9.2] proved that every monodromy operator, and hence every automorphism of X, acts on this group as $\pm\mathrm{id}$. Mongardi [Mon16, Lemma 24(4)] stated that any finite symplectic automorphism induces trivial action on the discriminant group of $H^2(X, \mathbb{Z})$. The argument relies on [Mon16, Lemma 23], but we have been unable to justify the assertion in its proof that $S_g(L_n)$ is negative definite (the subtlety is that g may have infinite order). This paper aims to provide an alternative proof of Mongardi's result:

Theorem 1.1. *Let X be a hyperkähler manifold of $K3^{[n]}$-type, and let $G \subset \mathrm{Aut}^s(X)$ be a finite subgroup. Then the induced action of G on $A_{H^2(X,\mathbb{Z})}$ is trivial. This implies that the induced action on $A_{S_G(H^2(X,\mathbb{Z}))}$ is also trivial.*

Definition 1.1. Let N be a hyperbolic lattice. We say that N satisfies property (W) if, for every primitive vector $v \in N$ with even square $v^2 \geq 4$, there exists $a \in N$ such that

$$a \notin \mathbb{Q}v, \qquad a^2 \geq -2, \qquad |(a, v)| \leq \frac{v^2}{2}. \tag{W}$$

Such an a is called a *short vector* for (N, v).

We write $\mathrm{II}_{p,q}$ for the even unimodular lattice of signature (p, q), where $p, q \geq 1$ and $p \equiv q \pmod 8$. For a single isometry g of a lattice L, write $S_g(L) = (L^g)^{\perp}_L$.

Theorem 1.2. *Let $F \in O(\mathrm{II}_{4,20})$ have finite order and fix a positive-definite three-dimensional subspace pointwise. If v is primitive, $v^2 = 2(n-1) \geq 4$, and $F(v) = -v$, then v has a short vector in $S_F(\mathrm{II}_{4,20})$.*

Proof. Write $\mathrm{ord}(F) = 2^k q$ with q odd. Since $F(v) = -v$, we have $k \geq 1$, $F^q(v) = -v$, and $S_{F^q}(\mathrm{II}_{4,20}) \subset S_F(\mathrm{II}_{4,20})$; replace F by F^q. If $k = 1$, Proposition 2.5 gives a solution of (W). Otherwise, suppose there is no short vector. Lemma 3.2 gives $\mathrm{ord}(F) \in \{4, 8\}$, contradicted by Propositions 3.10 and 3.13, respectively. □

Remark 1.3. For $n \geq 3$, Theorem 1.2 is equivalent to [Mon16, Lemma 23]. This follows from Definition 4.1.

A lattice L is 2-elementary if A_L has exponent at most two. For involutions of $\mathrm{II}_{4,20}$, the coinvariant lattice is even and 2-elementary. The Cohn–Elkies bounds [CE03] establish property (W) outside four exceptional pairs of invariants, which are treated using Nikulin's classification [Nik79] and Vinberg's reflection chambers [Vin72]. Our use of sphere-packing bounds is inspired by Giovenzana, Grossi, Onorati and Veniani's proof of symplectic rigidity for hyperkähler manifolds of OG10-type [GGOV24].

For higher two-power orders, we first use the classification of Leech involutions [GL11]. Unless this already produces a short vector, unimodular gluing gives an even unimodular lattice $K \simeq \mathrm{II}_{4,12}$ on which F induces an isometry g with $\mathrm{ord}(g) = \mathrm{ord}(F)/2$. We construct suitable vectors in K using the structure of 2-elementary lattices and lift them to short vectors in $\mathrm{II}_{4,20}$.

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AI disclosure. LLM-based AI tools were used in the course of this research to explore proof strategies, assist in the development and refinement of mathematical arguments, and improve the exposition. The authors take full responsibility for all results and arguments presented in this paper.

2. Order Two Case

2.1. **Sphere Packing Method.** We first recall the Cohn–Elkies sphere-packing bound. For $1 \le d \le 22$, let δ_d^{CE} denote the center-density bound for dimension d in [CE03, Table 3], increased by 10^{-5} to ensure a valid upper bound. Every packing of unit balls in $\mathbb{R}^d$ has center density at most δ_d^{CE}. In particular, if P is a Euclidean lattice of rank d with $\min(P) \ge 4$, where $\min(P) := \min\{(x,x) : 0 \ne x \in P\}$, then

$$\frac{1}{\sqrt{\det P}} \le \delta_d^{\mathrm{CE}}.$$

Inspired by the sphere-packing argument in [GGOV24, §§ 2.4 and 3.2], the following lemma uses the Cohn–Elkies bounds to study 2-elementary hyperbolic lattices.

For a finite abelian group A, let $\ell(A)$ denote its minimal number of generators.

Lemma 2.1. *Let N be an even hyperbolic 2-elementary lattice with $\mathrm{rk}(N) + \ell(A_N) \le 24$. If*

$$(\mathrm{rk}(N), \ell(A_N)) \notin \{(11,11),(12,12),(13,11),(14,10)\},$$

then N has property (W).

Proof. Put $r = \mathrm{rk}(N)$ and $a = \ell(A_N)$. If $r = 1$, then $N \simeq \langle 2 \rangle$, and property (W) holds vacuously. The hypothesis gives $r \le 24$. If $r = 24$, then $a = 0$ and N is even unimodular, which is impossible since its signature difference $1 - 23 = -22$ is not divisible by 8. Thus $2 \le r \le 23$.

Fix a primitive $v \in N$ with $v^2 = 2(n-1) \ge 4$, and suppose it has no short vector. Let P be the Euclidean lattice whose underlying group is the orthogonal projection of N onto $v^\perp$, equipped with the bilinear form on N rescaled by -1. Since v is primitive, it extends to a basis $v, e_2, \dots, e_{\mathrm{rk}(N)}$ of N. The projections of the e_i form a basis of the Euclidean lattice P, with Gram matrix

$$\left(\frac{(e_i,v)(e_j,v)}{v^2} - (e_i,e_j) \right)_{i,j \ge 2}.$$

Replacing each e_i by $e_i - \frac{(e_i,v)}{v^2} v$ makes the Gram matrix block diagonal without changing its determinant. Thus $|\det N| = v^2 \det P$, and

$$\det P = \frac{|\det N|}{v^2} = \frac{|A_N|}{v^2} = \frac{2^a}{2(n-1)}.$$

Let $0 \ne y \in P$ and choose a lift $x \in N$. Adding an integer multiple of v leaves its projection unchanged and changes (x,v) by multiples of $v^2 = 2(n-1)$, so we may arrange $|(x,v)| \le n-1$.

Since $y \neq 0$, we have $x \notin \mathbb{Q}v$. The assumption that there is no short vector gives $x^2 < -2$, hence $x^2 \leq -4$. Thus

$$\|y\|_P^2 = -x^2 + \frac{(x,v)^2}{v^2} \geq 4,$$

so $\min(P) \geq 4$. For distinct $x, y \in P$, we have $\|x-y\| \geq 2$. Thus balls of radius 1 centered at the lattice points have disjoint interiors. The lattice P has rank $r-1$ and covolume $\sqrt{\det P}$, so this packing has center density $1/\sqrt{\det P}$. The Cohn–Elkies bound in [CE03, Table 3] therefore gives

$$\sqrt{\frac{2(n-1)}{2^a}} = \frac{1}{\sqrt{\det P}} \leq \delta_{r-1}^{\mathrm{CE}}.$$

Squaring both sides yields $2(n-1)/2^a \leq (\delta_{r-1}^{\mathrm{CE}})^2$, and multiplying by 2^{a-1} gives

$$n - 1 \leq 2^{a-1}\big(\delta_{r-1}^{\mathrm{CE}}\big)^2.$$

Since $A_N \simeq (\mathbb{Z}/2\mathbb{Z})^a$, the Smith normal form of a Gram matrix of N has $r-a$ entries equal to 1 and a entries equal to 2. Its rank modulo 2 is therefore $r-a$. As N is even, the Gram matrix modulo 2 is alternating and has even rank, so $a \equiv r \pmod 2$. Outside the 4 excluded pairs, $a \leq \min\{r, 24-r\} - 2$ for $11 \leq r \leq 14$, while $a \leq \min\{r, 24-r\}$ otherwise. The bounds in [CE03, Table 3] give the following estimates for the right-hand side above, omitting the 4 excluded pairs:

range of r	maximizing (r,a)	$\delta_{r-1}^{\mathrm{CE}}$	bound on $n-1$
$2 \leq r \leq 10$	$(10,10)$	0.05901	< 1.79
$11 \leq r \leq 14$	$(12,10)$	0.05933	< 1.81
$15 \leq r \leq 21$	$(15,9)$	0.07751	< 1.54
22	$(22,2)$	0.37390	< 0.28
23	$(23,1)$	0.51232	< 0.27

This is a contradiction to $n - 1 \geq 2$. □

Remark 2.2. In [GGOV24, § 3.2], sphere-packing bounds are applied directly to the coinvariant lattice after reversing its sign. Here we adapt and refine this approach by using the projection lattice P.

Following the notation of the above proof, set $S = N \cap v^\perp$ and $t = \mathrm{div}_N(v)$. Write $L(m)$ for L with its bilinear form multiplied by a nonzero integer m. The projection lattice P contains $S(-1)$ with index v^2/t, and

$$|\det S| = \frac{v^2\, 2^a}{t^2}, \qquad \det P = \frac{2^a}{v^2}.$$

If v has no short vector, then $\min P \geq 4$ and $\min S(-1) \geq 4$. A sphere-packing bound on $S(-1)$ gives only a lower bound on v^2, while the same bound on P gives the upper bound needed for the contradiction.

Remark 2.3. Blichfeldt's bound [Bli29] gives instead

$$n - 1 \leq \frac{2^a \Gamma((\mathrm{rk}(N)+3)/2)^2}{2(2\pi)^{\mathrm{rk}(N)-1}}.$$

This already excludes the ranges $2 \le \mathrm{rk}(N) \le 9$ and $16 \le \mathrm{rk}(N) \le 23$, but leaves the intermediate ranks unresolved.

The Rogers–Leech bounds [Rog58], [Lee67] also prove Lemma 2.1. Using the values in [GGOV24, Table 1], increased by 10^{-4} to ensure a valid upper bound, the same projection estimate gives $n - 1 < 1.94$ outside the 4 exceptional pairs.

2.2. **Vinberg's Inequality.** We now recall Vinberg's reflection chambers for $I_{1,d}$, the odd unimodular lattice of signature $(1, d)$. Vinberg proves that these chambers have finite volume for $2 \le d \le 17$ [Vin72, § 4.1]. For $11 \le d \le 14$, let $\mathcal{C}_d \subset \mathbb{R}^{1,d}$ be the cone defined by the linear inequalities

$$a \ge 0, \qquad c_1 \ge \cdots \ge c_d \ge 0, \qquad a \ge c_1 + c_2 + c_3, \qquad 3a \ge \sum_{i=1}^{11} c_i,$$

and, when $d = 14$, also $4a \ge 2c_1 + \sum_{i=2}^{14} c_i$. Vinberg uses the quadratic form $-a^2 + \sum_{i=1}^{d} c_i^2$, whereas we use its negative. In our sign convention, the inequalities above are the wall inequalities of these chambers [Vin72, § 4.1, (56) and Table 4]. Since these chambers have finite volume, the cones defined by their wall inequalities lie in the closure of the positive cone [Vin72, § 1.1, (13)]. Thus

$$(a; c_1, \ldots, c_d) \in \mathcal{C}_d \quad \Longrightarrow \quad a^2 - \sum_{i=1}^{d} c_i^2 \ge 0. \tag{2.1}$$

Lemma 2.4. *Let N be an even hyperbolic 2-elementary lattice with $(\mathrm{rk}\, N, \ell(A_N)) \in \{(11, 11), (12, 12), (13, 11), (14, 10)\}$. Then N has property (W).*

Proof. The models [Nik79, Theorem 3.6.2] for the first three pairs are $I_{1,10}(2)$, $I_{1,11}(2)$, and $U \oplus A_1(-1)^{11}$; the last pair gives $U \oplus D_4(-1) \oplus E_8(-2)$ or $U \oplus D_4(-1) \oplus A_1(-1)^8$. Each model contains a full-rank sublattice $\mathcal{F} = I_{1,s}(2)$ with $s = \mathrm{rk}\, N - 1 \in \{10, 11, 12, 13\}$: this follows from $I_{1,1}(2) \subset U$, $A_1(-1)^4 \subset D_4(-1)$, and $\langle 2 \rangle \oplus E_8(-2) \simeq I_{1,8}(2)$. Since $\mathcal{F} \subset N \subset \mathcal{F}^\vee$, a primitive vector of square $2(n-1) \ge 4$ has coordinates

$$v = (a; b_1, \ldots, b_s), \qquad n - 1 = a^2 - \sum_{i=1}^{s} b_i^2 \in \mathbb{Z}_{\ge 2}, \qquad a, b_i \in \tfrac{1}{2}\mathbb{Z}.$$

Replace v by $-v$ if necessary, then change signs and permute the b_i to arrange $a > 0$ and $b_1 \ge \cdots \ge b_s \ge 0$. All sign changes and coordinate permutations are applied simultaneously to N and v. Set

$$u_1 = (1; 1, 1, 1, 0^{s-3}), \qquad u_2 = \begin{cases} (3; 1^{10}), & s = 10, \\ (3; 1^{11}, 0^{s-11}), & s \ge 11. \end{cases}$$

If $(v, u_j) < 0$ for $j = 1$ or 2, apply the reflection

$$s_{u_j}(x) = x - \frac{2(x, u_j)}{u_j^2} u_j.$$

Since u_j^2 is -2 or -4 and $(\mathcal{F}, u_j) \subset 2\mathbb{Z}$, these reflections preserve $\mathcal{F}$ and $\mathcal{F}^\vee$. They also preserve the component of the positive cone containing v, so the first coordinate of $s_{u_j}(v)$

remains positive. It is strictly smaller than a: the coefficient $2(v,u_j)/u_j^2$ and the first coordinate of u_j are both positive.

Apply each reflection to both N and v, and then apply sign changes and permutations again so that $b_1 \geq \cdots \geq b_s \geq 0$. These operations preserve $\mathcal{F} \subset N \subset \mathcal{F}^\vee$. The process terminates because a remains a positive half-integer and strictly decreases at each reflection [Con83], [Vin72]. At the end of this process, (v,u_1) and (v,u_2) are both nonnegative, which means

$$a \geq b_1 + b_2 + b_3, \qquad 3a \geq \sum_{i=1}^{\min(s,11)} b_i. \tag{2.2}$$

All transformations are isometries, so a short vector for the resulting pair pulls back to one for the original (N,v).

Suppose that v has no short vector. Consider the root $r = (0; 0^{s-1}, 1) \in \mathcal{F} \subset N$, then $(v,r) = -2b_s$. Then $2b_s \geq n$, hence $b_s \geq n/2$. Then adding $n/2$ after the last coordinate preserves the coordinate ordering:

$$\widehat{v} = (a; b_1, \ldots, b_s, \frac{n}{2}) \in \mathbb{R}^{1,s+1}.$$

For $s = 11, 12$, the ordering and (2.2) give $\widehat{v} \in \mathcal{C}_{s+1}$. For $s = 10$, the vector u_2 has square -2 and $(v,u_2) = 2(3a - \sum_{i=1}^{10} b_i) \geq 0$. Since there is no short vector, we have

$$3a - \sum_{i=1}^{10} b_i \geq \frac{n}{2},$$

which gives the remaining wall inequality for $\widehat{v} \in \mathcal{C}_{11}$.

For $s = 13$, take $u = (4; 2, 1^{12}) \in \mathcal{F} \subset N$. This vector is nonzero and has square 0, so $u \notin \mathbb{Q}v$ because $v^2 > 0$. It lies in the closure of the positive cone containing v, so $(v,u) > 0$. Since u is not short and (v,u) is a positive integer,

$$4a - 2b_1 - \sum_{i=2}^{13} b_i = \frac{(v,u)}{2} \geq \frac{n}{2},$$

which gives the remaining wall inequality for $\widehat{v} \in \mathcal{C}_{14}$. Thus $\widehat{v} \in \mathcal{C}_{s+1}$ in every case, and (2.1) implies a contradiction to

$$a^2 - \sum_{i=1}^{s} b_i^2 - \left(\frac{n}{2}\right)^2 = n - 1 - \left(\frac{n}{2}\right)^2 = -\frac{(n-2)^2}{4} < 0.$$

□

Proposition 2.5. *Let N be an even hyperbolic 2-elementary lattice with $\mathrm{rk}(N) + \ell(A_N) \leq 24$. Then N has property (W).*

In particular, if $F \in O(\mathrm{II}_{4,20})$ is an involution fixing a positive-definite three-space pointwise and $S_F(\mathrm{II}_{4,20})$ is hyperbolic, then $S_F(\mathrm{II}_{4,20})$ has property (W).

Proof. The first assertion follows from Lemmas 2.1 and 2.4. For the involution case, unimodular gluing makes $N = S_F(\mathrm{II}_{4,20})$ even and 2-elementary, with $\mathrm{rk}(N) + \ell(A_N) \leq 24$, so the first assertion applies. □

3. Higher Order Cases

3.1. **The Unimodular Reduction.** We first recall the Leech embedding lemma.

Lemma 3.1. *Let S be a positive-definite even lattice with no vectors of square 2, and let $G \subset O(S)$ be a finite subgroup acting trivially on A_S and satisfying $S^G = 0$. If $\operatorname{rk} S + \ell(A_S) \le 24$, then S admits a primitive embedding into the positive-definite Leech lattice $\mathbb{L}$. Moreover, the action of G extends to $\mathbb{L}$ and fixes $S_{\mathbb{L}}^{\perp}$ pointwise.*

This lemma was originally due to [GHV12, Appendix B.2]. For the primitivity issue in the original argument, see [MM25, Example 4.12 and Corollary 4.19] and [Zhe25, Lemma 1.1].

Throughout this section, let F and v satisfy the hypotheses of Theorem 1.2.

Lemma 3.2. *Assume $\operatorname{ord}(F) = 2^k$ with $k \ge 2$, let $N = S_F(\mathrm{II}_{4,20})$, and suppose (N, v) has no short vector. Set $H_0 = F^{2^{k-1}}$. Then $S_{H_0}(\mathrm{II}_{4,20}) \simeq E_8(-2)$ and $\operatorname{ord}(F) \in \{4, 8\}$.*

Proof. Write $C = S_{H_0}(\mathrm{II}_{4,20})$ and $R = C_{\mathrm{II}_{4,20}}^{\perp}$. Since $H_0(v) = v$, the fixed positive three-space together with v gives a positive four-space fixed by H_0. Thus C is negative definite, has rank at most 20, and lies in $N \cap v^{\perp}$. This lattice has no vector of square -2, since such a vector would satisfy (W). Moreover,

$$\ell(A_C) = \ell(A_R) \le \operatorname{rk} R = 24 - \operatorname{rk} C.$$

Since H_0 acts trivially on A_C, Lemma 3.1 applies to $(C(-1), \langle H_0 \rangle)$. The classification of Leech involutions [GL11, Appendix F, Theorem F.4] (see also [HM16, Table 1]) gives

$$C(-1) \simeq E_8(2), \quad BW_{16}, \quad D_{12}^{+}(2), \quad \mathbb{L}.$$

Leech lattice $\mathbb{L}$ itself is excluded by rank. For BW_16 and $D_{12}^{+}(2)$, $\operatorname{rk} R$ is respectively 8 and 12, and their discriminant groups satisfy

$$A_R \simeq A_C \simeq (\mathbb{Z}/2\mathbb{Z})^{\operatorname{rk} R}.$$

Since A_R is 2-elementary, $R^{\vee} \subset \frac{1}{2}R$. Both $R^{\vee}$ and $\frac{1}{2}R$ contain R with index $2^{\operatorname{rk} R}$, so $R^{\vee} = \frac{1}{2}R$. Hence $v/2 \in R^{\vee}$ because $v \in R$. Since v is primitive in $\mathrm{II}_{4,20}$, we have $v/2 \notin R$, so $v/2 + R$ is a nonzero discriminant class.

For $L = BW_{16}$ or $D_{12}^{+}(2)$, every $y \in L_{\mathbb{R}}$ admits $x \in L$ such that $(y - x)^2 \le 3$. For BW_{16}, rescale the bilinear form so that $\min BW_{16} = 4$; see [All00, proof of Theorem 3.1]. The covering radius of D_{12}^{+} is $\sqrt{12/8}$ [AE98, § III.B], hence the covering radius of $D_{12}^{+}(2)$ is $\sqrt{3}$. Hence every discriminant class of both lattices has a representative of square at most 3.

Choose $c \in C^{\vee}$ with $c^2 \ge -3$ such that $c + C$ corresponds to $v/2 + R$ under the gluing anti-isometry. Integral gluing gives $a = v/2 + c \in N$ with $(a, v) = n - 1$. Moreover,

$$a^2 \ge \frac{n-1}{2} - 3 \ge -2.$$

The class $v/2 + R$ is nonzero, so $a \notin \mathbb{Q}v$. Thus a is a short vector for (N, v), a contradiction to the standing assumption. Consequently $C \simeq E_8(-2)$.

Rescaling C gives an isometry $u \in O(E_8)$ with $u^{2^{k-1}} = -1$. Its eigenvalues are therefore primitive 2^k-th roots. Thus its characteristic polynomial is a power of the cyclotomic polynomial $\Phi_{2^k}(t)$, whose degree is 2^{k-1}. Since $\operatorname{rk} E_8 = 8$, we have $2^{k-1} \mid 8$. The only possibilities are $k = 2, 3, 4$.

Suppose $k = 4$. Then the characteristic polynomial is $t^8 + 1$. Put $T = 1 - u$; we obtain

$$[E_8 : TE_8] = \det T = 2.$$

Choose $x \in E_8 \setminus TE_8$. Since E_8/TE_8 has order 2, there is $z \in E_8$ with $Tz = 2x$. If $z = 2y$ for some $y \in E_8$, then $Tz = 2x$ gives $x = Ty \in TE_8$, a contradiction. Thus $z \notin 2E_8$.

For every $y \in E_8$, the isometry u gives

$$(Ty, z) = (y, (1 - u^{-1})z) = -2(y, u^{-1}x) \in 2\mathbb{Z}.$$

Thus TE_8 is contained in the kernel of the homomorphism $E_8 \to \mathbb{Z}/2\mathbb{Z}$, $y \mapsto (y, z) \pmod 2$. This homomorphism is nonzero: otherwise $z/2 \in E_8^\vee = E_8$, a contradiction to $z \notin 2E_8$. Its kernel therefore has index 2. Since TE_8 also has index two, it equals this kernel. Our choice $x \notin TE_8$ therefore forces (x, z) to be odd.

On the other hand, $uz = z - 2x$ and preservation of the square give

$$z^2 = (uz)^2 = (z - 2x)^2 = z^2 - 4(x, z) + 4x^2.$$

Hence $(x, z) = x^2$, which is even because E_8 is even. This contradiction excludes $k = 4$, so $\operatorname{ord}(F) = 4$ or 8. □

For the rest of this section, we follow the assumptions and notation of Lemma 3.2. Recall that

$$C = S_{H_0}(\mathrm{II}_{4,20}) \simeq E_8(-2), \qquad R = C^\perp_{\mathrm{II}_{4,20}}.$$

Definition 3.1. Consider $\gamma : A_C \to A_R$, the anti-isometry induced by unimodular gluing. The isometry F induces an action on A_C, which we also denote by F, and put

$$W = \{F^{\operatorname{ord}(F)/4}(\alpha) - \alpha : \alpha \in A_C\}.$$

Let K be the overlattice of R defined by

$$K = \{x \in R^\vee : x + R \in \gamma(W)\}.$$

Proposition 3.3. *The lattice K of Definition 3.1 is an F-invariant even unimodular overlattice of R:*

$$R \subset K \subset R^\vee, \qquad [K : R] = 16, \qquad K \simeq \mathrm{II}_{4,12}. \tag{3.1}$$

Let $g = F|_K$. Then $\operatorname{ord}(g) = \operatorname{ord}(F)/2$, and on A_R the involution $h = g^{\operatorname{ord}(F)/4}$ satisfies

$$\operatorname{im}(h - 1) = \ker(h - 1) = K/R.$$

Proof. Let $d = \operatorname{ord}(F)/2$. We have a natural identification

$$C/2C \cong A_C, \qquad c + 2C \mapsto c/2 + C.$$

Let E be the underlying group of C with form $(x, y)_E = -\frac{1}{2}(x, y)_C$; then $E \simeq E_8$. Let $u = F|_E$ and $w = u^{d/2}$. Since $w^2 = -1$,

$$(w - 1)^2 = -2w.$$

Hence $2E \subset (w - 1)E$, and taking determinants gives $|\det(w - 1)| = 2^4$. Under $C/2C = E/2E$, we have $W = (w - 1)E/2E$, so

$$|W| = [(w - 1)E : 2E] = \frac{[E : 2E]}{[E : (w - 1)E]} = \frac{2^8}{2^4} = 2^4.$$

Next, $w^{-1} = -w$ implies $(wx, x)_E = 0$ for every $x \in E$. Thus $((w-1)x)_E^2 = 2x_E^2$, and the discriminant quadratic form satisfies

$$q_C((w-1)x + 2E) = -\frac{((w-1)x)_E^2}{2} = -x_E^2 \equiv 0 \pmod{2\mathbb{Z}}.$$

Here the last congruence relies on the evenness of E. Therefore, W is totally isotropic. Since γ is an anti-isometry, $\gamma(W)$ is also totally isotropic, and its inverse image $K \subset R^\vee$ is even. Moreover, $K/R \cong \gamma(W)$ gives $[K : R] = 2^4$. Since $|A_R| = |A_C| = 2^8$,

$$|\det K| = \frac{|\det R|}{[K : R]^2} = \frac{2^8}{(2^4)^2} = 1.$$

Thus K is unimodular. Its signature is $(4, 12)$, so $K \simeq \mathrm{II}_{4,12}$, giving (3.1).

Since W is F-invariant and γ is F-equivariant, K is F-invariant. On $E/2E$, the identity $(w-1)^2 = 0$ implies $\mathrm{im}(w-1) \subset \ker(w-1)$. Both have dimension 4, so $\mathrm{im}(w-1) = \ker(w-1) = W$. Transporting this equality by γ gives $\mathrm{im}(h-1) = \ker(h-1) = K/R$.

Since $u^d = -1 \equiv 1 \pmod 2$ but $u^{d/2} = w \not\equiv 1 \pmod 2$, the action of u on A_C is of order d. The same holds on A_R, while $F^d = H_0$ fixes R and hence K. Thus $\mathrm{ord}(g) = d$. □

Lemma 3.4. *If*

$$\text{(P)} \qquad b \in S_g(K), \qquad b^2 \geq 0, \qquad 0 < (b, v) \leq n - 1,$$

then (N, v) has a short vector.

Proof. Identify $C = E_8(-2)$, with the roots of E_8 having square 2. Choose $x \in E_8$ so that the class of $x/2 \in C^\vee$ maps to $b + R \in A_R$ under the gluing anti-isometry. The covering radius of E_8 is 1 in this normalization [CS99, Ch. 21, § 3.D]; hence some $y \in E_8$ satisfies $\|x/2 - y\|^2 \leq 1$. Thus $c = (x - 2y)/2$ represents the same class and has

$$c^2 = -\frac{1}{2}\|x - 2y\|^2 \geq -2.$$

Then $a = b + c \in \mathrm{II}_{4,20}$ has square at least -2 and $(a, v) = (b, v)$. Since C has no F-fixed vectors, $(K^g)_\mathbb{Q} = (\mathrm{II}_{4,20}{}^F)_\mathbb{Q}$, so $a \in N$. The primitivity of v in $\mathrm{II}_{4,20}$ implies $\mathbb{Q}v \cap \mathrm{II}_{4,20} = \mathbb{Z}v$; as $0 < (a, v) \leq n - 1 < v^2$, we have $a \notin \mathbb{Q}v$. □

The following lemma is a natural continuation of Proposition 3.3. We will use it repeatedly to rule out $\mathrm{ord}(F) = 4$ and 8.

Lemma 3.5. *Suppose an involution $h \in O(K)$ preserves R and satisfies, on A_R,*

$$\mathrm{im}(h-1) = \ker(h-1) = K/R.$$

Let $K_- = K^{-h}$ and $K_+ = K^h$. Then

$$\mathrm{rk}\, K_- - \ell(A_{K_-}) \geq 4, \qquad \mathrm{rk}\, K_- + \ell(A_{K_-}) \leq 12.$$

The map $K_- \to K/R$ is surjective with kernel $K_- \cap R$ containing $2K_-^\vee$. The kernel of the induced map $K_-/2K_-^\vee \to K/R$ is nondegenerate for the alternating pairing $(x, y) \pmod 2$. For each of the lattices K_+ and K_-, $\mathrm{rk}\, K_\pm - \ell(A_{K_\pm}) = 4$ implies $K_\pm \cap R = 2K_\pm^\vee$.

Proof. The space $V = 2R^\vee/2K \subset K/2K$ is four-dimensional and totally isotropic. The inverse image of its annihilator with respect to the bilinear form in K is R. For $w, w' \in 2R^\vee$, set

$$B(\overline{w}, \overline{w}') = \left(\frac{(h-1)w}{2}, w'\right) \pmod 2.$$

The isomorphism $(h-1) : R^\vee/K \to K/R$ and the perfect pairing $K/R \times V \to \mathbb{F}_2$ induced by $(\cdot,\cdot) \pmod 2$ show that B is well-defined and nondegenerate. Writing $w = 2x$, we have $(h-1)x \in K$ and obtain $B(\overline{w}, \overline{w}) = -((h-1)x)^2 = 0 \pmod 2$.

Since $(h-1)(w/2) \in K$, the vectors $(w \pm hw)/2$ belong to $K_\pm$, respectively. Replacing w by $w + 2k$, with $k \in K$, adds $k \pm hk$ to $(w \pm hw)/2$. Since $(k \pm hk)/2 \in K_\pm^\vee$, the resulting classes modulo $2K_\pm^\vee$ are unchanged. Thus there are well-defined maps

$$j_\pm : V \longrightarrow K_\pm/2K_\pm^\vee, \qquad \overline{w} \longmapsto \frac{w \pm hw}{2} + 2K_\pm^\vee.$$

Their induced forms are B and $-B$, respectively. Since B is nondegenerate, both maps are injective. Unimodular gluing makes $K_\pm$ even and 2-elementary, with equal discriminant lengths. The quotients $K_\pm/2K_\pm^\vee$ therefore have nondegenerate alternating pairings and are of dimension $\operatorname{rk} K_\pm - \ell(A_{K_\pm})$. The embeddings give $\operatorname{rk} K_\pm - \ell(A_{K_\pm}) \geq 4$. Since $\ell(A_{K_+}) = \ell(A_{K_-})$ and $\operatorname{rk} K_+ + \operatorname{rk} K_- = 16$, we obtain

$$\operatorname{rk} K_- + \ell(A_{K_-}) = 16 - \big(\operatorname{rk} K_+ - \ell(A_{K_+})\big) \leq 12.$$

The same argument gives $\operatorname{rk} K_+ + \ell(A_{K_+}) \leq 12$.

For $z \in K_\pm$ and $w \in 2R^\vee$, we have $\big(z, \frac{w \pm hw}{2}\big) = (z, w)$. The nondegenerate pairing between K/R and V identifies K/R with $\operatorname{Hom}(V, \mathbb{F}_2)$. Under this identification, the map $K_\pm \to K/R$ sends x to $\xi \mapsto (x, j_\pm(\xi)) \bmod 2$. It factors through $K_\pm/2K_\pm^\vee$ and is surjective. Its kernel is the orthogonal complement of the nondegenerate subspace $j_\pm(V)$, hence is nondegenerate. When the quotient has dimension 4, this kernel is zero, giving $K_\pm \cap R = 2K_\pm^\vee$. □

3.2. **Nonnegative Short Vectors.** The preceding rank bound leads to the following nonnegative short-vector theorem.

Theorem 3.6. *Let L be an even 2-elementary lattice of signature $(1, r-1)$, with $r + \ell(A_L) \leq 12$. For every $v \in L$ with $v^2 = 2(n-1) \geq 4$, there exists $b \in L$ with $b^2 \geq 0$ and $0 < (v, b) \leq n-1$.*

Proof. If $L_0 = L \oplus D$ with D negative definite and $b \in L_0$ satisfies the conclusion for $v \in L$, then its projection $b_L \in L$ satisfies $(v, b_L) = (v, b)$ and $b_L^2 \geq b^2$.

Let $a = \ell(A_L)$. Since the Gram form of an even lattice is alternating modulo 2, $r - a$ is even. Thus

$$k = \frac{12 - r - a}{2} \in \mathbb{Z}_{\geq 0}, \qquad L^\sharp = L \oplus A_1(-1)^k, \qquad \operatorname{rk} L^\sharp + \ell(A_{L^\sharp}) = 12.$$

By the projection argument, we may therefore assume $r + a = 12$. For $x \in L^\vee$ and $\overline{x} = x + L$, the discriminant quadratic form is $q_L(\overline{x}) = x^2 \pmod{2\mathbb{Z}}$. Put $\delta = 0$ if q_L is integer-valued and $\delta = 1$ otherwise. Since $a \leq r$, we have $6 \leq r \leq 12$; the case $r = 12$, $a = 0$ is excluded by the signature condition for even unimodular lattices. If $\delta = 0$, the existence conditions

in [Nik79, Theorem 3.6.2] leave only $(r,a)=(10,2)$. Thus the block decomposition [Ale25, § 9.2] and indefinite uniqueness [Nik79, Theorem 3.6.2] give exactly 7 triples:

$$(6,6,1),\ (7,5,1),\ (8,4,1),\ (9,3,1),\ (10,2,0),\ (10,2,1),\ (11,1,1).$$

They are represented, in order, by $I_{1,5}(2)$, $U\oplus A_1(-1)^5$, and $U\oplus Q(-1)$ for

$$Q = D_4\oplus A_1^2,\ D_6\oplus A_1,\ D_8,\ E_7\oplus A_1,\ E_8\oplus A_1.$$

Lemmas 3.8 and 3.9 below prove the assertion for these models. □

We verify the models using Voronoi cells and two explicit reductions.

Definition 3.2. For a positive-definite lattice Q, the Voronoi cell centered at 0 is defined by the nearest-point condition [CS99, Ch. 2, § 1.2]:

$$\mathrm{Vor}(Q)=\{x\in Q\otimes_{\mathbb{Z}}\mathbb{R} : \|x\|\le\|x-q\| \text{ for all } q\in Q\}.$$

Lemma 3.7. *Let Q be a positive-definite even lattice. Suppose $\mathrm{Vor}(Q)$ is contained in the convex hull of finitely many vectors h, each of which either has $h^2\le 3/2$ or has $h^2=2$ with $h\in Q^\vee$ and $2h\in Q$. For every $v\in U\oplus Q(-1)$ with $v^2=2(n-1)\ge 4$, there exists $b\in U\oplus Q(-1)$ with $b^2\ge 0$ and $0<(v,b)\le n-1$.*

Proof. Use $\|z\|^2$ and $(z,h)_Q$ for the norm and pairing in the positive-definite lattice Q. Choose a hyperbolic basis e,f of U so that $v=\alpha e+\beta f+z$ with $\alpha,\beta>0$ and

$$n-1=\alpha\beta-\frac{1}{2}\|z\|^2.$$

For $t\in Q$, the Eichler translation fixing f is the isometry of $U\oplus Q(-1)$ given by

$$\alpha e+\beta f+z\longmapsto \alpha e+\left(\beta+(z,t)_Q+\frac{\alpha}{2}\|t\|^2\right)f+z+\alpha t,$$

where $(\ ,\ )_Q$ is the pairing on Q. Choosing t appropriately, we may assume $z/\alpha\in\mathrm{Vor}(Q)$. Exchanging e,f whenever $\beta<\alpha$ strictly decreases the positive integer α. Thus we may assume $\beta\ge\alpha$ and $z/\alpha\in\mathrm{Vor}(Q)$. If $\alpha\le n-1$, take $b=f$. Otherwise $\alpha>n-1\ge 2$. The norm bound $\|z\|^2\le 2\alpha^2$ gives $n-1\ge\alpha(\beta-\alpha)$, hence $\beta=\alpha$.

Because z/α lies in this convex hull, one of the specified vectors h satisfies $(z,h)_Q\ge\|z\|^2/\alpha$. If $h^2\le 3/2$, Cauchy's inequality gives $\|z\|^2\le 3\alpha^2/2$ and $n-1\ge\alpha^2/4$. For integers $\alpha\ge 3$, this contradicts $n-1<\alpha$, using integrality when $\alpha=3$.

Otherwise $h^2=2$. Since $(z,h)_Q\le\sqrt{2}\|z\|<2\alpha$, the integer $\delta=2\alpha-(z,h)_Q$ is positive. The support inequality gives $n-1\ge\alpha\delta/2$, so $\delta=1$. The integral vector $b=2e+2f+2h\in U\oplus Q(-1)$ is isotropic and has $(v,b)=2\delta=2\le n-1$. □

Lemma 3.8. *Let Q be any of the lattices*

$$D_4\oplus A_1^2,\quad D_6\oplus A_1,\quad D_8,\quad E_7\oplus A_1,\quad E_8\oplus A_1.$$

For every $v\in U\oplus Q(-1)$ with $v^2=2(n-1)\ge 4$, there exists $b\in U\oplus Q(-1)$ with $b^2\ge 0$ and $0<(v,b)\le n-1$.

Proof. For an irreducible root lattice of rank j, the part of its Voronoi cell in a dominant Weyl chamber satisfies $(x,\theta)\le 1$, where $\theta=\sum_{i=1}^{j}c_i\alpha_i$ is the highest root [CS99, Ch. 21, Theorem 5]. It is therefore contained in the simplex with vertices $0,\omega_i/c_i$, where ω_i are

the fundamental weights. The images of these vertices under the Weyl group, and their Cartesian products for direct sums, give the required finite set.

The nonzero A_1 vertex has square $1/2$. For D_j, the spinor vertices $\frac{1}{2}(1,\dots,1,\pm1)$ have square $j/4$; all other vertices have square at most $\max(1,(j-2)/4)$. In the numbering with the E_7 chain $1,\dots,6$ and vertex 7 attached to 3, the highest-root coefficients $(2,3,4,3,2,1,2)$ give one vertex ω_6 of square $3/2$; all the others have square at most 1.

The squared covering radius of E_8 is one [CS99, Ch. 21, § 3.D], so every Voronoi vertex of $E_8 \oplus A_1$ has square at most $3/2$. For the other 4 lattices, every product vertex has square at most $3/2$, except for vertices of square 2 formed from a square-one D_4 vertex and two A_1 vertices, a D_6 spinor vertex and one A_1 vertex, a D_8 spinor vertex, or ω_6 and one A_1 vertex. All these vertices lie in $Q^\vee$, and their doubles lie in Q: this holds for the A_1 vertices, the relevant D_4 vertices, the spinor weights of even D_j, and ω_6 since $|A_{E_7}| = 2$. Lemma 3.7 applies. □

Lemma 3.9. *Let L be one of the lattices*

$$U \oplus A_1(-1)^5, \qquad I_{1,5}(2).$$

For every $v \in L$ with $v^2 = 2(n-1) \geq 4$, there exists $b \in L$ with $b^2 \geq 0$ and $0 < (v,b) \leq n-1$.

Proof. For $U \oplus A_1(-1)^5$, write

$$v = \alpha e + \beta f + \sum_{i=1}^{5} c_i r_i, \qquad r_i^2 = -2.$$

Eichler transformations, sign changes, and exchanges give $0 < \alpha \leq \beta$ and $0 \leq c_i \leq \alpha/2$. Also arrange

$$p = \alpha + \beta - \sum_{i=1}^{5} c_i \geq 0$$

by reflecting in $2e + 2f + \sum_{i=1}^{5} r_i$, a root. If $p < 0$, this replaces α, β by $\alpha + 4p, \beta + 4p$ and decreases their positive sum; Reducing each c_i modulo α to $[-\alpha/2, \alpha/2]$ by Eichler translations keeps α fixed and does not increase β. Hence the process terminates.

If $\alpha \leq n-1$, use f. Otherwise, let $t = \min_i c_i$. Since $t \leq c_i \leq \alpha/2$,

$$c_i^2 \leq \left(\frac{\alpha}{2} + t\right) c_i - \frac{\alpha t}{2}.$$

Summing and using $\sum_{i=1}^{5} c_i = \alpha + \beta - p$ gives

$$n - 1 \geq \frac{\alpha}{2}(p+t) + (\beta - \alpha)\left(\frac{\alpha}{2} - t\right) + pt \geq \frac{\alpha}{2}(p+t).$$

As $n-1 < \alpha$, the nonnegative integer $p+t$ is at most 1. If $c_{i_0} = t$, then $b = 2e + 2f + \sum_{\substack{i=1 \\ i \neq i_0}}^{5} r_i$ is a nonzero isotropic vector with $(v,b) = 2(p+t)$. The orthogonal complement of the positive vector v is negative definite, so $p + t = 1$ and $(v,b) = 2 \leq n-1$.

Finally take $I_{1,5}(2)$ with integral coordinates $v = (a; b_1, \dots, b_5)$, $a > 0$, and $b_1 \geq \cdots \geq b_5 \geq 0$. Reflect in $(1;1,1,1,0,0)$ to arrange $a \geq b_1 + b_2 + b_3$. Let $p = a - b_1 > 0$, so $p \geq b_2 + b_3$. Then

$$n - 1 \geq p^2 + 2pb_2 - b_2^2 - 3b_3^2 \geq p^2,$$

because $2pb_2 - b_2^2 - 3b_3^2 \geq (b_2 - b_3)(b_2 + 3b_3) \geq 0$. The isotropic vector $u = (1; 1, 0, \ldots, 0)$ satisfies $(v, u) = 2p \leq n - 1$: for $p = 1$ this follows from $n - 1 \geq 2$, while for $p \geq 2$ we have $2p \leq p^2 \leq n - 1$. □

3.3. **Order Four And Eight Cases.**

Proposition 3.10. *If (N, v) has no short vector, then* $\operatorname{ord}(F) \neq 4$.

Proof. If $\operatorname{ord}(F) = 4$, then g is an involution and $S_g(K) = K^{-g}$ is even, hyperbolic, and 2-elementary. Lemma 3.5 gives $\operatorname{rk} S_g(K) + \ell(A_{S_g(K)}) \leq 12$. Theorem 3.6 supplies (P), a contradiction to Lemma 3.4. □

It remains to exclude $\operatorname{ord}(F) = 8$.

Lemma 3.11. *In the setting of Proposition 3.3, suppose* $\operatorname{ord}(F) = 8$, *so* $\operatorname{ord}(g) = 4$. *Let* $C' = K^{-g^2}$, *and write* $q = \operatorname{rk} C'$ *and* $a = \ell(A_{C'})$. *Then* $C' \subset S_g(K) \cap v^\perp$ *is negative definite,*

$$q - a \geq 4, \qquad q + a \leq 12,$$

and the quotient map $C' \to K/R$ *is surjective and annihilates* $2(C')^\vee$. *Moreover,*

$$C'(-1) \simeq D_6,\ D_{10},\ \text{or } D_4^2.$$

Proof. The assertions about signature and orthogonality follow from $g^2(v) = v$ and the fixed positive three-space. The involution g^2 is the one in Proposition 3.3. Lemma 3.5 gives the inequalities and the quotient map. Any root in $C' \cap R \subset N \cap v^\perp$ would satisfy (W), so $C' \cap R$ is rootless. Since the quotient map annihilates $2(C')^\vee$, a root of divisibility two would belong to $C' \cap R$. Thus C' has no such roots.

Since $g^2 = -1$ on C', its rank q is even. As C' is even and 2-elementary, $q - a$ is even, so a is also even. Combining the two inequalities with the existence conditions for even 2-elementary forms [Nik79] and [FK15, Theorem 3.6.3] gives

$$(q, a) = (6, 2), (8, 0), (8, 2), (8, 4), (10, 2).$$

Set $E = C'(-1)$ and identify their underlying groups. For $(q, a) = (6, 2)$, the finite quadratic form is $\langle -1/2 \rangle^2$. Gluing with A_1^2 gives E_8. Orthogonal pairs of roots in E_8 are conjugate and have complement D_6, so $E = D_6$.

Since E is positive definite of rank 8, its discriminant form has Brown invariant (signature modulo 8) equal to 0 [Nik79, Theorem 1.3.3]. Finite quadratic-form classification [FK15, Theorem 3.6.3] gives a totally isotropic subgroup of order $2^{a/2}$; gluing along it produces an even unimodular overlattice $\Lambda \simeq E_8$ of index $2^{a/2}$. Since E is 2-elementary, $2\Lambda \subset E$. Let $V = \Lambda/2\Lambda$ with $Q(x) = x^2/2 \pmod 2$, and let $Z = (E/2\Lambda)^\perp$. Then $\dim Z = a/2$, and

$$Z = 2E^\vee/2\Lambda \subset Z^\perp = E/2\Lambda, \qquad E/2E^\vee \simeq Z^\perp/Z.$$

By Lemma 3.5,

$$2\Lambda \subset 2E^\vee \subset E \cap R.$$

Every class of V with $Q = 1$ has a root representative: the squared covering radius of E_8 is 1, so each class has a representative of square at most 4. Since $E \cap R$ is rootless and contains 2Λ, Q vanishes on $(E \cap R)/2\Lambda$. In particular, it vanishes on Z and descends to $E/2E^\vee$. Its restriction to

$$H = (E \cap R)/2E^\vee = \ker(E/2E^\vee \to K/R)$$

is therefore zero. Lemma 3.5 makes the polarization on H nondegenerate, so $H = 0$. Since $\dim H = 4 - a$, we obtain $a = 4$ and Z is a totally singular plane. Root reflections realize $O(V, Q)$ [CS99]; by Witt's lemma, Z is equivalent to the plane spanned by $2e_1$ and $e_1 + e_2 + e_3 + e_4$ in standard coordinates. The orthogonal preimage $E \subset \Lambda$ is $D_4 \oplus D_4$: orthogonality to $2e_1$ removes the half-integral vectors, while orthogonality to $e_1 + e_2 + e_3 + e_4$ imposes even sums in each four-coordinate block.

For $(q, a) = (10, 2)$, the discriminant form is $\langle 1/2 \rangle^2$. Gluing with D_6 produces an even unimodular rank-sixteen lattice, either E_8^2 or D_{16}^+ [CS99]. In the former, the connected D_6 root system lies in one E_8 factor and has complement $E_8 \oplus A_1^2$. In the latter case, after permuting coordinates, the roots of this D_6 subsystem are $\pm e_i \pm e_j$ with $1 \le i < j \le 6$, where $e_1, \dots, e_{16}$ is the standard orthonormal basis. Orthogonality forces these coordinates to vanish, which excludes every vector of $D_{16}^+ \setminus D_{16}$, whose coordinates all lie in $\mathbb{Z} + \frac{1}{2}$. The remaining 10 coordinates are integers with even sum, so the complement is D_{10}. □

Proposition 3.12. *Let K be even and unimodular of signature $(4, 12)$. Let $g \in O(K)$ have order 4 and fix a positive-definite three-space pointwise. If $K^{-g^2} \simeq D_6(-1)$, then every $v \in K$ with $g(v) = -v$ and $v^2 = 2(n-1) \ge 4$ admits $b \in S_g(K)$ with $b^2 \ge 0$ and $0 < (v, b) \le n - 1$.*

Proof. Let $C' = K^{-g^2}$ and $R_1 = (C')^{\perp}_K$. Every isometry of D_6 is a signed permutation. Since $(g|_{C'})^2 = -1$, its permutation consists of 3 two-cycles, each with negative sign product. Thus g exchanges the two spinor classes of $A_{C'}$ and hence those of A_{R_1}.

Let τ be the sign change of the first coordinate in $D_6(-1)$. It exchanges the spinor classes $\frac{1}{2}(1,1,1,1,1,\pm 1) + C'$ and fixes their sum, so τ and $g|_{C'}$ induce the same action on $A_{C'}$. Since $g^2 = 1$ on R_1, unimodular gluing extends $h = (g|_{R_1}) \oplus \tau$ to an involution of K. Let $L = K^{-h}$, $r = \operatorname{rk} L$, and $a = \ell(A_L) = \ell(A_{K^h})$. Unimodular gluing makes both eigensublattices even and 2-elementary.

The fixed positive three-space lies in K^h. Moreover $v \in L$, while $K^g \subset K^h$ because g has no fixed vectors in C'. Hence L is hyperbolic and $L \subset S_g(K)$.

Inside $D_6(-1)$, the coordinate vectors supported on positions $2, \dots, 6$ and summing to zero form $A_4(-1) \subset K^h$. Its determinant is 5, so the Gram matrix of K^h has rank at least 4 modulo 2. Since K^h is 2-elementary, this rank equals $\operatorname{rk}(K^h) - a = 16 - r - a$. Thus $r + a \le 12$. Theorem 3.6 applied to $v \in L$ gives $b \in L \subset S_g(K)$ satisfying (P). □

Proposition 3.13. *If (N, v) has no short vector, then $\operatorname{ord}(F) \ne 8$.*

Proof. Suppose $\operatorname{ord}(F) = 8$, and use the notation of Lemma 3.11. If $C' = D_6(-1)$, Proposition 3.12 gives (P), a contradiction to Lemma 3.4.

The remaining possibilities are $C' = D_{10}(-1)$ and $D_4(-1)^2$, both with $q + a = 12$. Lemma 3.5 gives $\operatorname{rk}(K^{g^2}) - \ell(A_{K^{g^2}}) = 16 - q - a = 4$ and therefore $K^{g^2} \cap R = 2(K^{g^2})^\vee$. Since $v \in K^{g^2} \cap R$, we have $v/2 \in (K^{g^2})^\vee$. Every discriminant class of $D_4(-1)^2$ has a representative of square at least -2; for $D_{10}(-1)$ the corresponding bound is $-5/2$, attained by the spinor classes. Choose a representative $c \in (C')^\vee$ corresponding to the class of $v/2$. Then $b = v/2 + c \in K$ and $(b, v) = n - 1$. Its square satisfies

$$b^2 \ge \frac{n-1}{2} - \frac{5}{2} \ge -\frac{3}{2} > -2.$$

Evenness of K gives $b^2 \geq 0$. Both summands are perpendicular to K^g, so (P) holds, a contradiction to Lemma 3.4. □

4. Proof of The Main Theorem

Let X be a hyperkähler manifold of $K3^{[n]}$-type, $n \geq 2$, and choose a representative of its canonical Mukai embedding orbit, so that $H = H^2(X, \mathbb{Z}) = v^\perp \subset \mathrm{II}_{4,20}$ and $v^2 = 2(n-1)$. Write $\mathcal{K}_X$ for its Kähler cone.

Definition 4.1. For a primitive class $D \in \mathrm{Pic}(X)$ with $D^2 < 0$, put

$$T_D = \mathrm{II}_{4,20} \cap \mathrm{span}_{\mathbb{Q}}\{v, D\}.$$

The class D is a numerical wall divisor if there exists $a \in T_D \setminus \mathbb{Q}v$ such that

$$a^2 \geq -2, \qquad 0 \leq (a, v) \leq v^2/2.$$

Equivalently, there exists $a \in T_D \setminus \mathbb{Q}v$ satisfying either

$$a^2 = -2, \quad 0 \leq (a, v) \leq v^2/2, \qquad \text{or} \qquad 0 \leq a^2 v^2 < (a, v)^2 \leq (v^2/2)^2.$$

The second condition is from [Mon16, Definition 9]. Notice that [Mon16, Definition 9] contains a typo: the last two inequality signs, $\leq$ and $<$, should be interchanged.

The series work about wall classes in [BM14], [BHT15, Theorem 1], [Mon15, Theorems 1.3 and 2.4], and [Mon16, Theorem 10] tells that every numerical wall divisor D satisfies $D^\perp \cap \mathcal{K}_X = \varnothing$.

Proof of Theorem 1.1. Average a Kähler class to obtain a G-invariant class $\kappa \in \mathcal{K}_X$. The action on A_H is automatically trivial for $n = 1, 2$. For $n \geq 3$, suppose that some $f \in G$ acts nontrivially on A_H. Markman's monodromy description [Mar11, Lemma 9.2] gives $f^*|_{A_H} = -\mathrm{id}$. Unimodular gluing in the canonical embedding [Mar11, Corollary 9.5(1)] extends f^* and $v \mapsto -v$ to a finite-order $F \in O(\mathrm{II}_{4,20})$. Since f is symplectic, F fixes the positive three-space spanned by $\kappa, \mathrm{Re}\,\sigma, \mathrm{Im}\,\sigma$. Theorem 1.2 therefore gives a short vector $a \in S_F(\mathrm{II}_{4,20})$ for v; namely, after choosing its sign,

$$a \notin \mathbb{Q}v, \qquad a^2 \geq -2, \qquad 0 \leq (a, v) \leq v^2/2.$$

The lattice $S_f(H)$ is negative definite and algebraic, and $S_F(\mathrm{II}_{4,20})_{\mathbb{Q}} = \mathbb{Q}v \oplus S_f(H)_{\mathbb{Q}}$. Thus the projection

$$c = a - \frac{(a, v)}{v^2} v \in S_f(H)_{\mathbb{Q}} \setminus \{0\}$$

is algebraic and has negative square. Choose a primitive class $D \in S_f(H)$ spanning $\mathbb{Q}c$. Moreover, $a \in T_D \setminus \mathbb{Q}v$. Definition 4.1 shows that D is a numerical wall divisor. But κ is invariant, so $(D, \kappa) = 0$. Hence

$$\kappa \in D^\perp \cap \mathcal{K}_X,$$

a contradiction to the wall-divisor property. Thus G acts trivially on A_H.

For the second assertion, when $n \geq 2$, extend the G-action to $\mathrm{II}_{4,20}$ fixing v, using its trivial action on A_H; the coinvariant lattice is unchanged. Unimodular gluing then gives triviality on $A_{S_G(H)}$, and for $n = 1$ the same argument applies directly to the unimodular lattice H. □

References

[AE98] E. Agrell and T. Eriksson, *Optimization of lattices for quantization*, IEEE Trans. Inform. Theory **44** (1998), no. 5, 1814–1828.

[Ale25] V. Alexeev, *Reflective hyperbolic 2-elementary lattices, K3 surfaces and hyperkahler manifolds*, Higher dimensional algebraic geometry—a volume in honor of V. V. Shokurov, London Math. Soc. Lecture Note Ser., vol. 489, Cambridge Univ. Press, Cambridge, 2025, pp. 130–150.

[All00] D. Allcock, *New complex- and quaternion-hyperbolic reflection groups*, Duke Math. J. **103** (2000), no. 2, 303–333.

[BHT15] A. Bayer, B. Hassett, and Y. Tschinkel, *Mori cones of holomorphic symplectic varieties of K3 type*, Ann. Sci. Éc. Norm. Supér. (4) **48** (2015), no. 4, 941–950.

[BM14] A. Bayer and E. Macrì, *MMP for moduli of sheaves on K3s via wall-crossing: nef and movable cones, Lagrangian fibrations*, Invent. Math. **198** (2014), no. 3, 505–590.

[Bli29] H. F. Blichfeldt, *The minimum value of quadratic forms, and the closest packing of spheres*, Math. Ann. **101** (1929), no. 1, 605–608.

[CE03] H. Cohn and N. Elkies, *New upper bounds on sphere packings. I*, Ann. of Math. (2) **157** (2003), no. 2, 689–714.

[Con83] J. H. Conway, *The automorphism group of the* 26*-dimensional even unimodular Lorentzian lattice*, J. Algebra **80** (1983), no. 1, 159–163.

[CS99] J. H. Conway and N. J. A. Sloane, *Sphere packings, lattices and groups*, third ed., Grundlehren der mathematischen Wissenschaften, vol. 290, Springer-Verlag, New York, 1999.

[FK15] S. Finashin and V. Kharlamov, *Apparent contours of nonsingular real cubic surfaces*, Trans. Amer. Math. Soc. **367** (2015), no. 10, 7221–7289.

[FWZ26] J. Fu, S. Wang, and Z. Zheng, *A Proof of Mongardi's Conjecture on Finite Symplectic Group Actions on Hyperkähler Manifolds of* $K3^{[n]}$ *Type*, to appear on arXiv soon.

[GGOV24] L. Giovenzana, A. Grossi, C. Onorati, and D. C. Veniani, *Symplectic rigidity of O'Grady's tenfolds*, Proc. Amer. Math. Soc. **152** (2024), no. 7, 2813–2820.

[GHV12] M. R. Gaberdiel, S. Hohenegger, and R. Volpato, *Symmetries of K3 sigma models*, Commun. Number Theory Phys. **6** (2012), no. 1, 1–50.

[GL11] R. L. Griess, Jr. and C. H. Lam, EE_8*-lattices and dihedral groups*, Pure Appl. Math. Q. **7** (2011), no. 3, 621–743.

[HM16] G. Höhn and G. Mason, *The 290 fixed-point sublattices of the Leech lattice*, J. Algebra **448** (2016), 618–637.

[Lee67] J. Leech, *Notes on sphere packings*, Canadian J. Math. **19** (1967), 251–267.

[Mar11] E. Markman, *A survey of Torelli and monodromy results for holomorphic-symplectic varieties*, Complex and differential geometry, Springer Proc. Math., vol. 8, Springer, Heidelberg, 2011, pp. 257–322.

[MM25] L. Marquand and S. Muller, *Finite groups of symplectic birational transformations of IHS manifolds of OG*10 *type*, Forum Math. Sigma **13** (2025), Paper No. e117, 40.

[Mon15] G. Mongardi, *A note on the Kähler and Mori cones of hyperkähler manifolds*, Asian J. Math. **19** (2015), no. 4, 583–591.

[Mon16] ———, *Towards a classification of symplectic automorphisms on manifolds of* $K3^{[n]}$ *type*, Math. Z. **282** (2016), no. 3-4, 651–662.

[Nik79] V. V. Nikulin, *Integer symmetric bilinear forms and some of their geometric applications*, Izv. Akad. Nauk SSSR Ser. Mat. **43** (1979), no. 1, 111–177, 238.

[Rog58] C. A. Rogers, *The packing of equal spheres*, Proc. London Math. Soc. (3) **8** (1958), 609–620.

[Vin72] È. B. Vinberg, *The groups of units of certain quadratic forms*, Mat. Sb. (N.S.) **87(129)** (1972), 18–36.

[Zhe25] Z. Zheng, *A lemma on Leech-like lattices*, 2025, arXiv:2507.10414v2.

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